Is NYC Housing and Age a Great Predictor of Income?

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Introduction

Living in one of the most expensive cities in the world, housing takes up a large share of many New Yorkers' living expense. According to the 2023 NYC Housing and Vacancy Survey, the median renter household spent 29.5% of its income on housing costs. Therefore, household income can make a huge difference in the housing people can afford in New York City, but how closely is income related to someone's age, their housing conditions, and when they moved to the city? This project explores how income is related to the respondent's age, the number of maintenance problems in their home, and the year they moved to NYC. We analyze these relationships to better predict a household's income based on these characteristics.

Exploratory Data Analysis

Data

Our dataset contains information on a random sample of 299 households in New York City according to New York City Housing and Vacancy Survey. We explore how household income varies with the respondent's age, the number of maintenance deficiencies in their home, and the year they moved to NYC. The four variables used in this analysis are the following:

`Income`: the household's total income (in $)

`Age`: age of the respondent (in years)

`MaintenanceDef`: count of the residence's maintenance deficiencies during the 2002-2005 period.

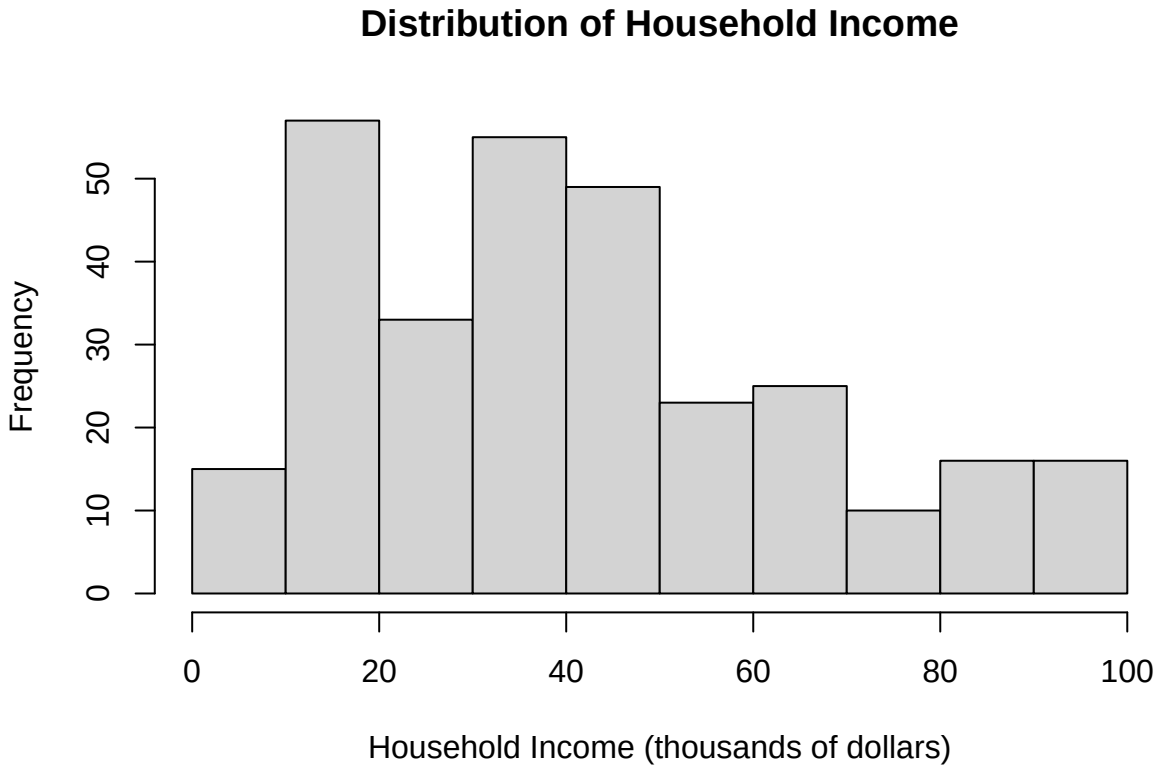
`NYCMove`: the year in which the respondent moved to New York City

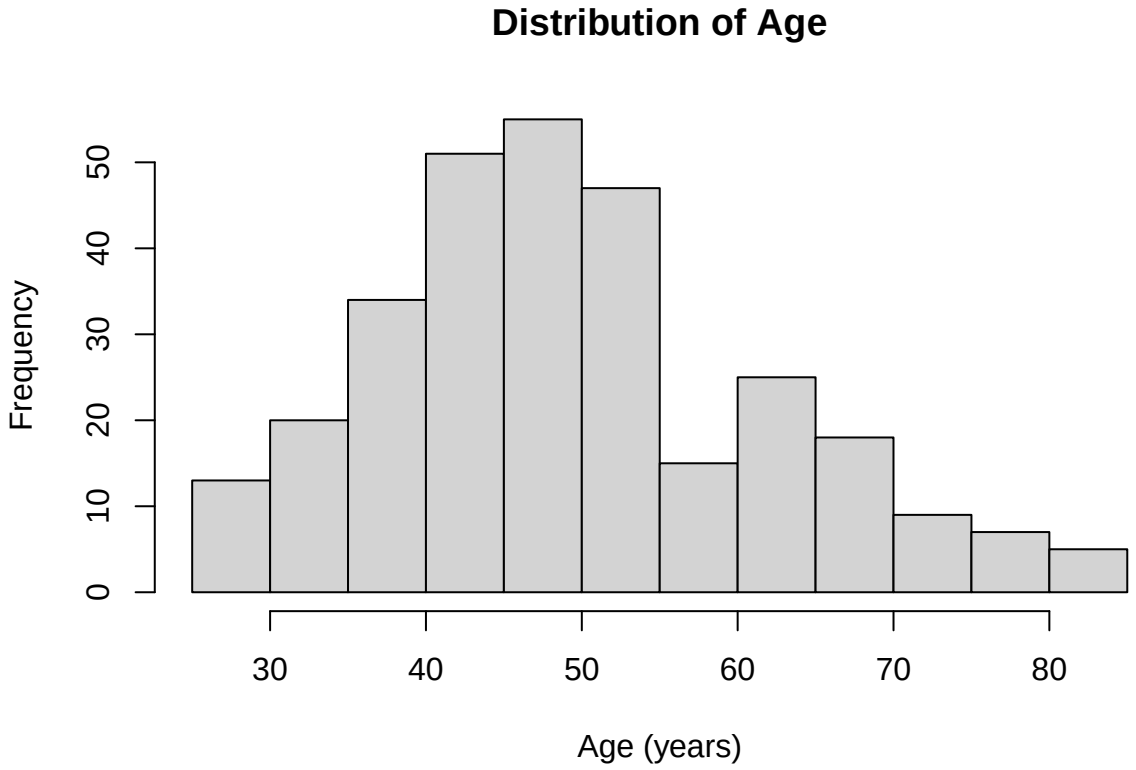
Here are the first few lines of data:

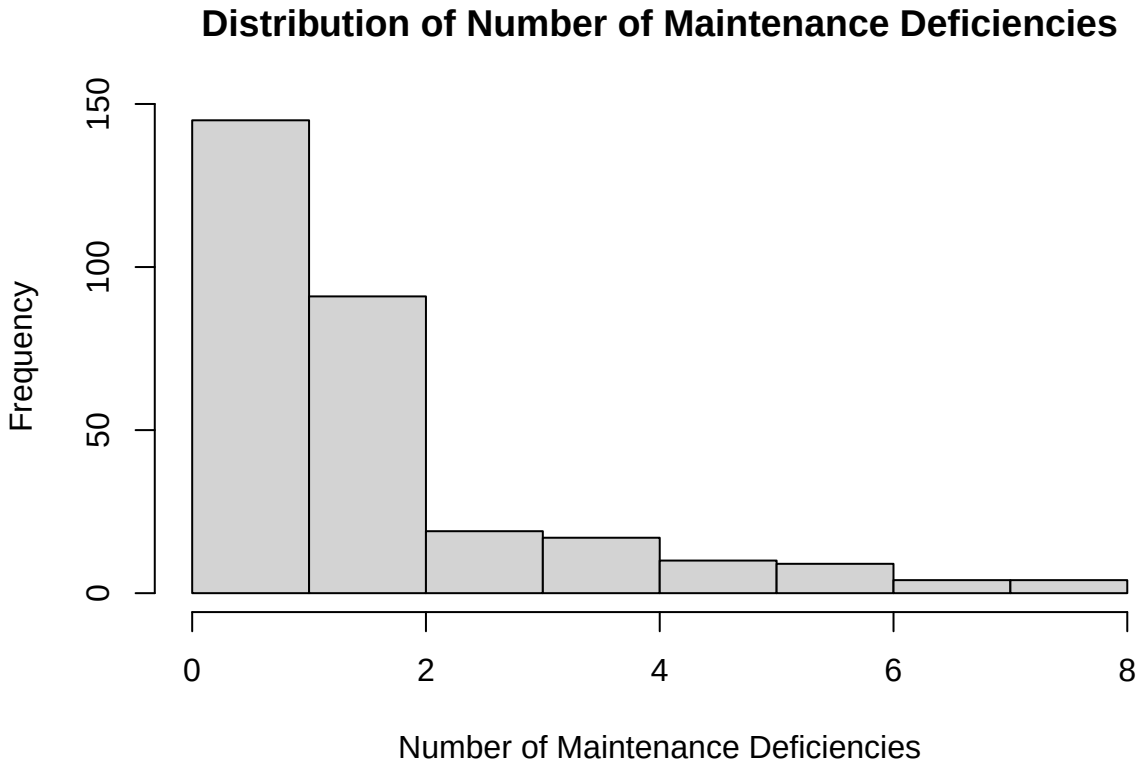
```
## # A tibble: 6 x 4
##   Income   Age MaintenanceDef NYCMove
##    <dbl> <dbl>          <dbl>   <dbl>
## 1   8400    77              1    1981
## 2  17510    53              2    1986
## 3  19200    33              4    1992
## 4  42717    55              1    1969
## 5   5000    58              2    1989
## 6  30000    29              4    1994
```

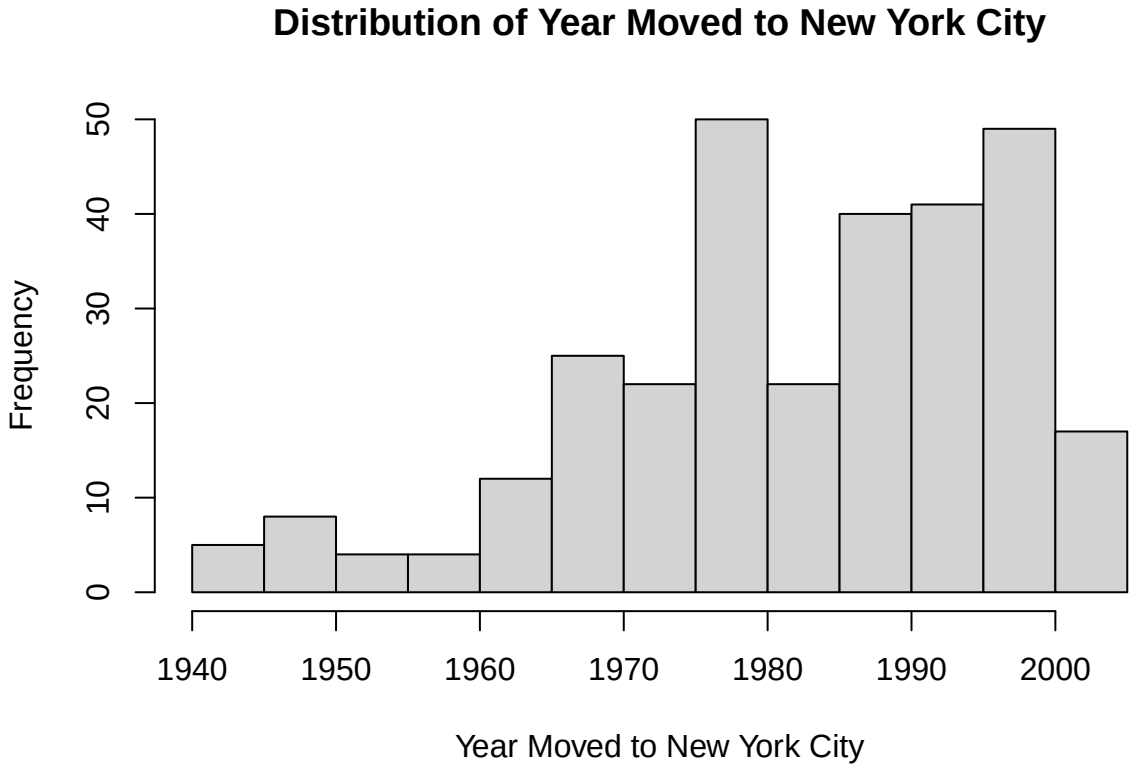
Univariate exploration

Since all variables are quantitative, we use histograms to explore the distributions.




Distribution of Age
Frequency
0
10
20
30
40
50
30
40
50
60
70
80
Age (years)


Distribution of Number of Maintenance Deficiencies
Frequency
0
50
100
150
0
2
4
6
8
Number of Maintenance Deficiencies



The following are the univariate numerical summaries of the variables:

For `Income`:

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1440   21000   39000   42266   57800   98000
```

For `Age`:

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   26.00   42.00   49.00   50.03   58.00   85.00
```

For `MaintenanceDef`:

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.00    1.00    2.00    1.98    2.00    8.00
```

For `NYCMove`:

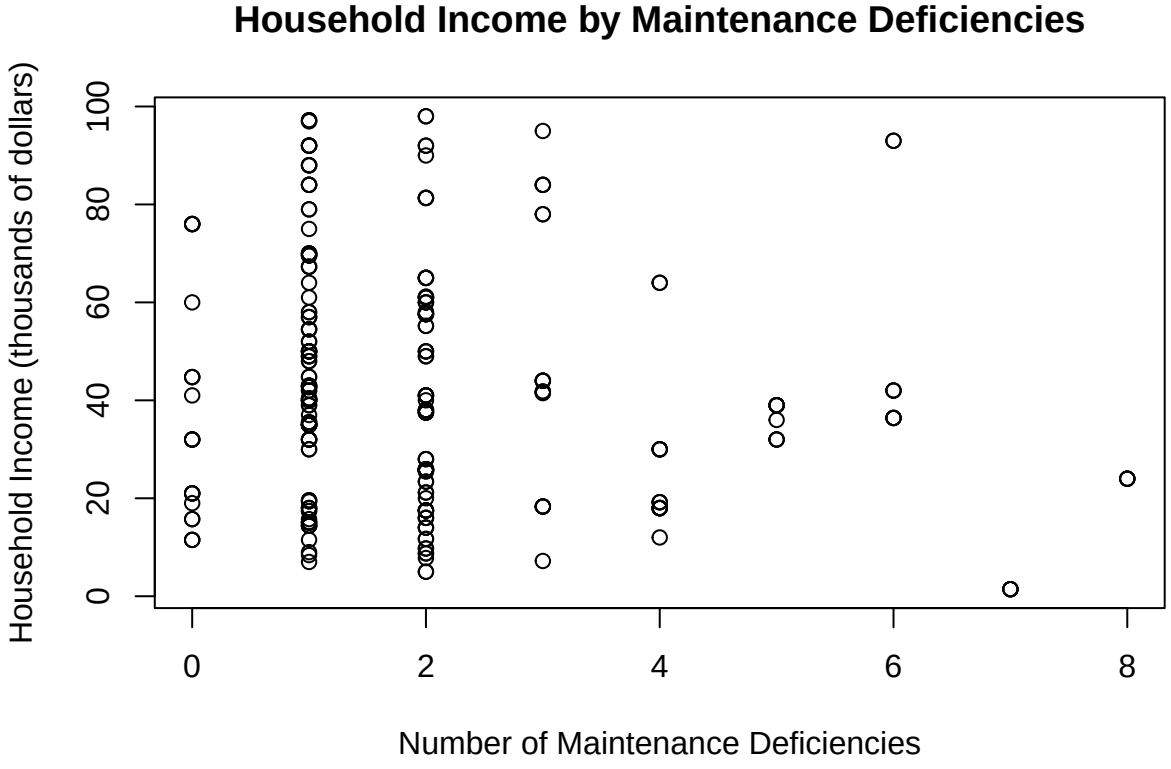
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1942    1973    1985    1983    1995    2004
```

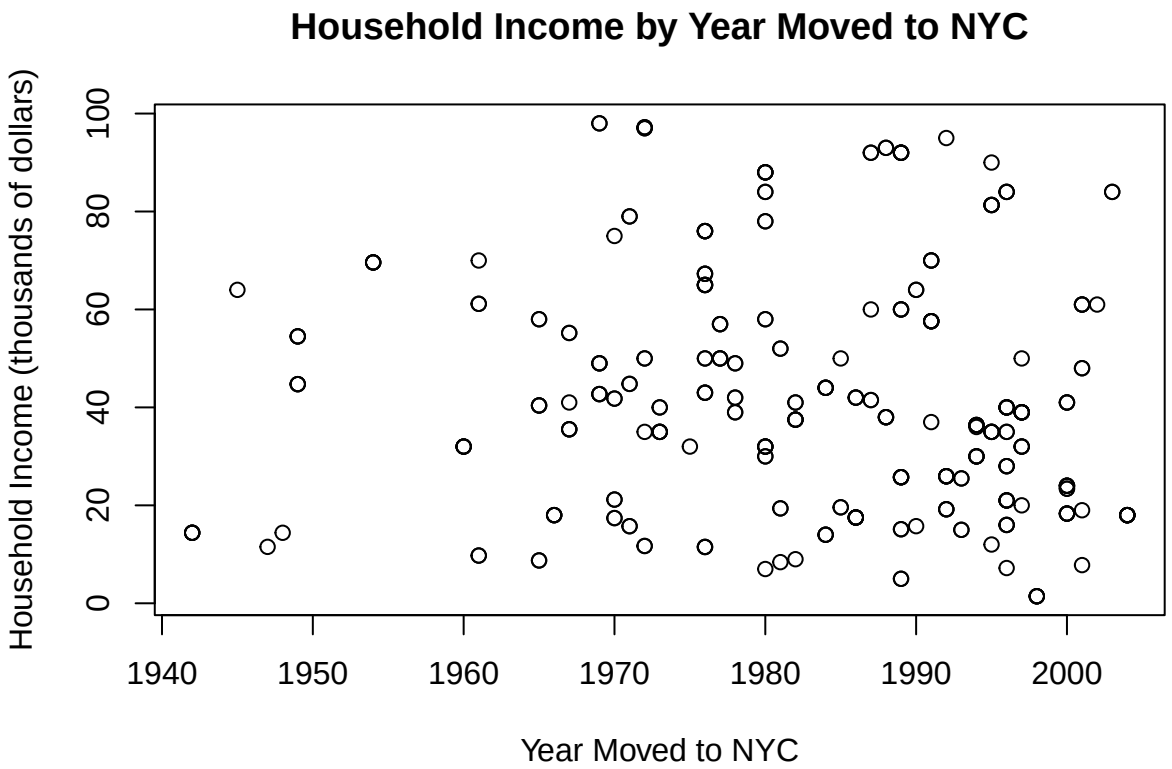
We make the following observations about both the graphs and the summary statistics. The distribution of respondents' total household **Income** is unimodal and skewed to the right. The distribution is centered around 39000 with the standard deviation of around 24000 and IQR of 36800. The distribution of **Age** is roughly symmetric and slightly bimodal. Most respondents are aged between 40 and 60. The distribution of number of household **Maintenance Deficiencies** is skewed to the right with some possible outliers of 7 or 8 maintenance deficiencies. Most households have less than or equal to 2 maintenance deficiencies. The distribution of **Year Moved to NYC** is skewed to the left and bimodal, with two spikes at 1975s and 1990s. Most responsdents moved to NYC between 1973 and 1995.

Bivariate Exploration

After understanding the distribution of the individual variables in this dataset, we can graphically investigate how each predictor is associated with the response variable, `Household Income`. Since all four variables involved in this study are quantitative, we use scatterplots to analyze the relationship between each explanatory variable and the response.



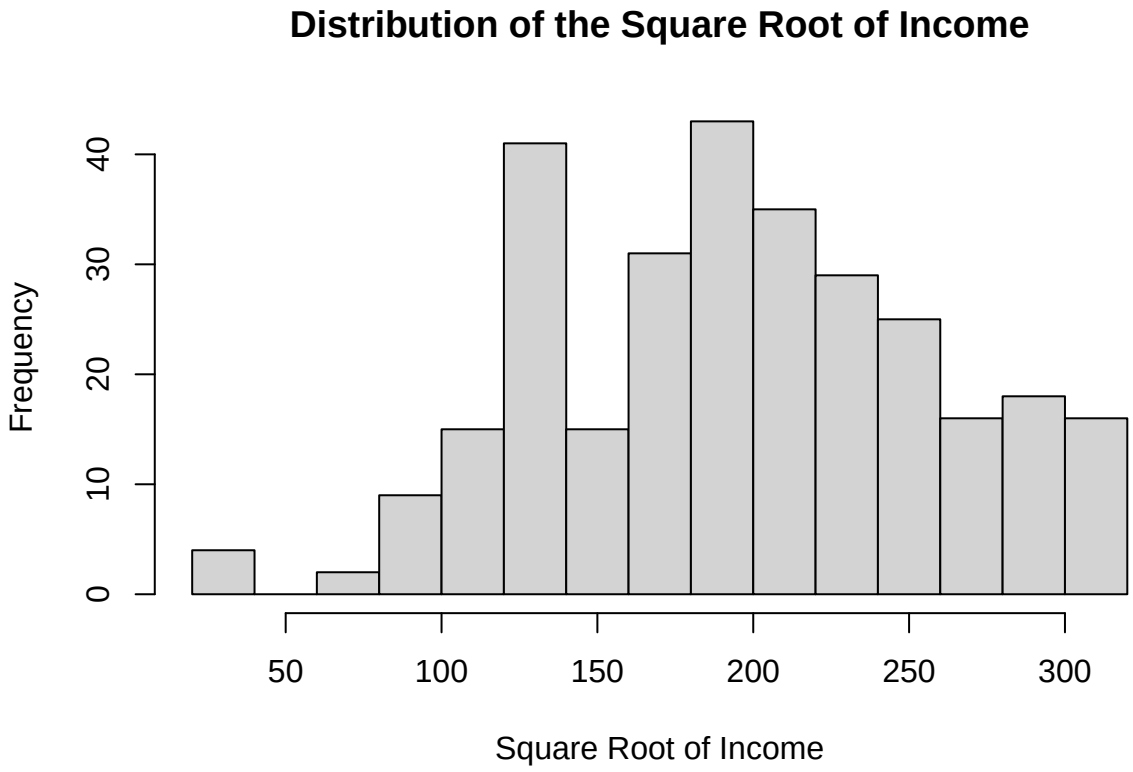




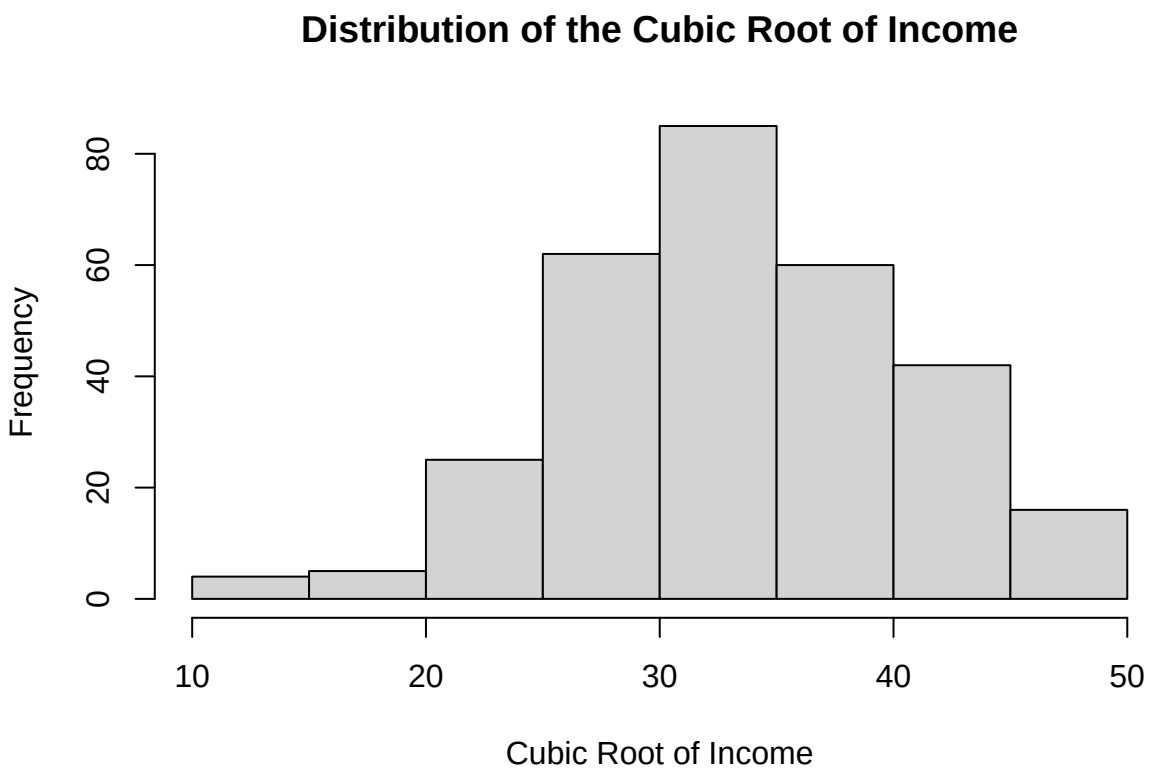
The scatterplot of household income by **age** shows a weak curved relationship. Higher incomes appear more common among respondents aged between 40 and 60, while younger and older respondents generally have lower incomes. However, the pattern is unclear because income varies widely at similar age level. The scatterplot of household income against **maintenance deficiencies** shows no clear increasing or decreasing trend. Households with one or two deficiencies have a wide range of incomes. We also don't have enough samples of household with more than 3 deficiencies, making the pattern difficult to tell. The scatterplot of household income against **year moved to NYC** shows no strong linear relationship as well. Respondents who came to NYC at similar time have very different household incomes. Overall, none of the three predictors shows a strong relationship with household income individually, so we want to consider them together in a regression model.

Modeling

Now we want to build a linear regression model to predict household income. We start by looking at the histogram of our response variable. It is skewed to the right, so we want to apply some transformation to `household income` variable. We start by applying square root:



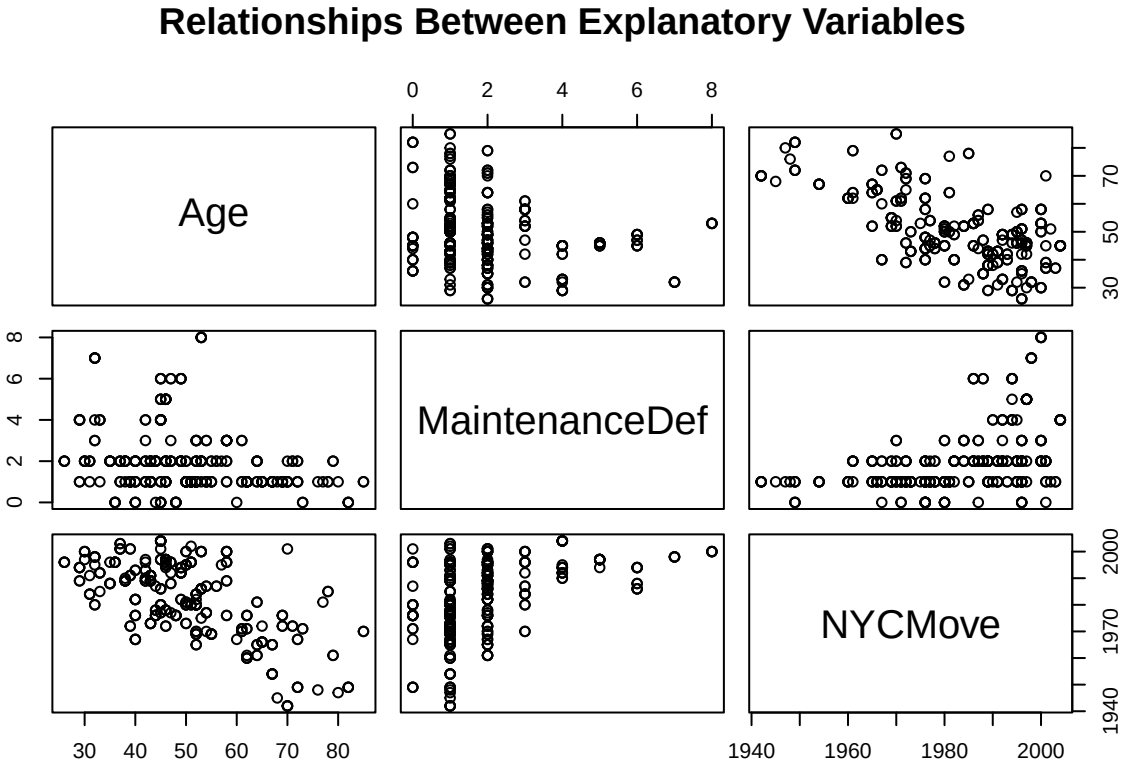
We can see it's still bimodal with a fat tail to the greater end, so we continue to apply cubic root on `household income`.



Now it looks approximately symmetrical. We can try applying cube root transformation on response variable, `Income`, in our final model.

Checking multicollinearity

Now we want to consider all three explanatory variables together. First, we explore the relationships between the explanatory variables and check their VIFs:




```
##             Age MaintenanceDef         NYCMove
##        1.687649       1.267728        1.999724
```

Although there seems to be a negative relationship between `Age` and `NYCMove`, the VIFs are approximately 1.6876, 1.2677, and 1.9997. Since all three are below 2.5, there is no strong concern about multicollinearity in this model.

Transformation on an explanatory variable

We saw in the bivariate exploration that the relationship between household income and age may not be a straight line. Therefore, we want to try applying a log transformation to `Age`. This transformation allows the association with income to change more quickly at younger ages and more slowly at older ages. However, `LogAge` by itself cannot describe income increasing and then decreasing with age.

```
##
## Call:
## lm(formula = IncomeCubRt ~ Age + MaintenanceDef + NYCMove, data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.8251  -5.0337   0.3266   5.1024  15.4301
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    85.33627   82.38862   1.036  0.30115
## Age            -0.02307    0.04282  -0.539  0.59040
## MaintenanceDef -0.85738    0.28494  -3.009  0.00285 **
```

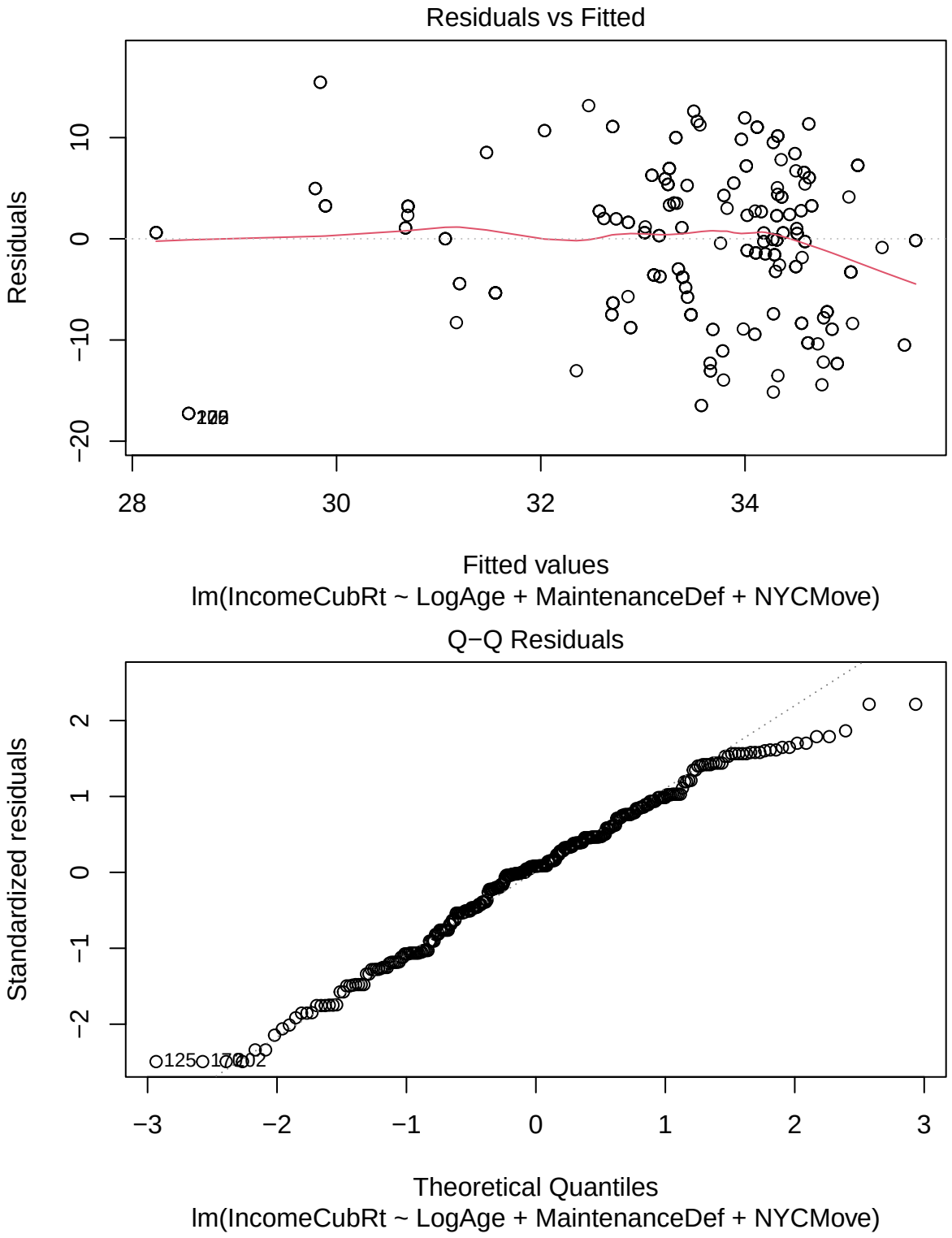
```
## NYCMove        -0.02476    0.04100  -0.604  0.54634
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.076 on 295 degrees of freedom
## Multiple R-squared:  0.04344,    Adjusted R-squared:  0.03371
## F-statistic: 4.466 on 3 and 295 DF,  p-value: 0.004376
```

```
##
## Call:
## lm(formula = IncomeCubRt ~ LogAge + MaintenanceDef + NYCMove,
##     data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.2588  -5.0822   0.5838   5.2755  15.4657
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)    29.2431616 83.7300350   0.349  0.72715
## LogAge          1.0999189  2.0768387   0.530  0.59678
## MaintenanceDef -0.8749558  0.2848419  -3.072  0.00233 **
## NYCMove         0.0008112  0.0398688   0.020  0.98378
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.077 on 295 degrees of freedom
## Multiple R-squared:  0.04341,    Adjusted R-squared:  0.03368
## F-statistic: 4.462 on 3 and 295 DF,  p-value: 0.004396
```

```
##         LogAge MaintenanceDef        NYCMove
##       1.586923       1.266776       1.890506
```

We can compare from the summaries that the log transformation does not improve the fit by itself. The adjusted R-squared is about 0.03368, compared with 0.03371 for the model using original age. In the log age model, the p-values for `LogAge` and `NYCMove` are about 0.597 and 0.984, which are way above 0.05. Only `MaintenanceDef` is significant at the 5% significance level, with a p-value of about 0.00233. The VIFs are all below 2.5 after the transformation as well, suggesting no multicollinearity concerns.

Then, we use the following residual plot and Q-Q plot to check the regression assumptions:



The points on the residual plot scatter around zero, but the red line bends downward at the upper end. This suggests that the model may not capture the mean relationship. There is no clear funnel shape, although the lack of observations at lower fitted values leaves constant variance a concern. The points on the Q-Q plot follow the straight line closely in the middle but deviate at both ends, especially the upper end. Therefore, the Normality condition is only approximately satisfied. Applying log to age alone does not resolve these concerns.

Checking an interaction

Although there are a few potentially insignificant variables, before removing variables, we also want to consider whether the relationship between maintenance deficiencies and household income depends on age. We check an interaction between `LogAge` and `MaintenanceDef`.

We include both explanatory variables and their interaction directly in the model.

```
##
## Call:
## lm(formula = IncomeCubRt ~ LogAge + MaintenanceDef + LogAge:MaintenanceDef +
##     NYCMove, data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.5937  -4.7319   0.7432   4.5001  13.9907
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            170.79606   88.27070   1.935  0.05396 .
## LogAge                  -9.34070    3.21638  -2.904  0.00396 **
## MaintenanceDef         -20.17944    4.63415  -4.355 1.84e-05 ***
## NYCMove                 -0.05037    0.04070  -1.238  0.21678
## LogAge:MaintenanceDef    5.09516    1.22093   4.173 3.96e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.888 on 294 degrees of freedom
## Multiple R-squared:  0.09691,    Adjusted R-squared:  0.08462
## F-statistic: 7.887 on 4 and 294 DF,  p-value: 4.742e-06
```

```
##                LogAge        MaintenanceDef               NYCMove
##              4.017942            353.957142              2.079384
## LogAge:MaintenanceDef
##            348.982600
```

The p-value for the interaction term is about 0.0000396, which is below 0.05. Therefore, there is evidence that the association between maintenance deficiencies and cube root of household income depends on the respondent's age. The adjusted R-squared increases to about 0.08462 and the residual standard error decreases to about 6.888. However, the VIFs for `LogAge`, `MaintenanceDef`, and their interaction are approximately 4.02, 353.96, and 348.98, while the VIF for `NYCMove` is about 2.08. The interaction term is strongly related to the variables used to calculate it. Therefore, multicollinearity becomes a concern after adding the interaction, even though it was not a strong concern in the model without interaction. We keep the interaction because it is significant and improves the fit, but we need to consider the high VIFs as a limitation of the model.

Removing a nonsignificant variable

We can see from the interaction model summary that `NYCMove` is not significant because its p-value is about 0.217. Therefore, we try removing it and fit the following simpler model:

```
##
## Call:
## lm(formula = IncomeCubRt ~ LogAge + MaintenanceDef + LogAge:MaintenanceDef,
##     data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -17.1222  -5.2859   0.6897   4.8051  14.4467
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)               62.272     10.251   6.075 3.82e-09 ***
## LogAge                    -7.033      2.623  -2.681  0.00775 **
## MaintenanceDef           -18.586      4.456  -4.171 3.99e-05 ***
## LogAge:MaintenanceDef      4.640      1.165   3.982 8.62e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.894 on 295 degrees of freedom
## Multiple R-squared:  0.0922, Adjusted R-squared:  0.08297
## F-statistic: 9.987 on 3 and 295 DF,  p-value: 2.74e-06
```

```
##                 LogAge        MaintenanceDef LogAge:MaintenanceDef
##               2.667872            326.639031            317.283256
```

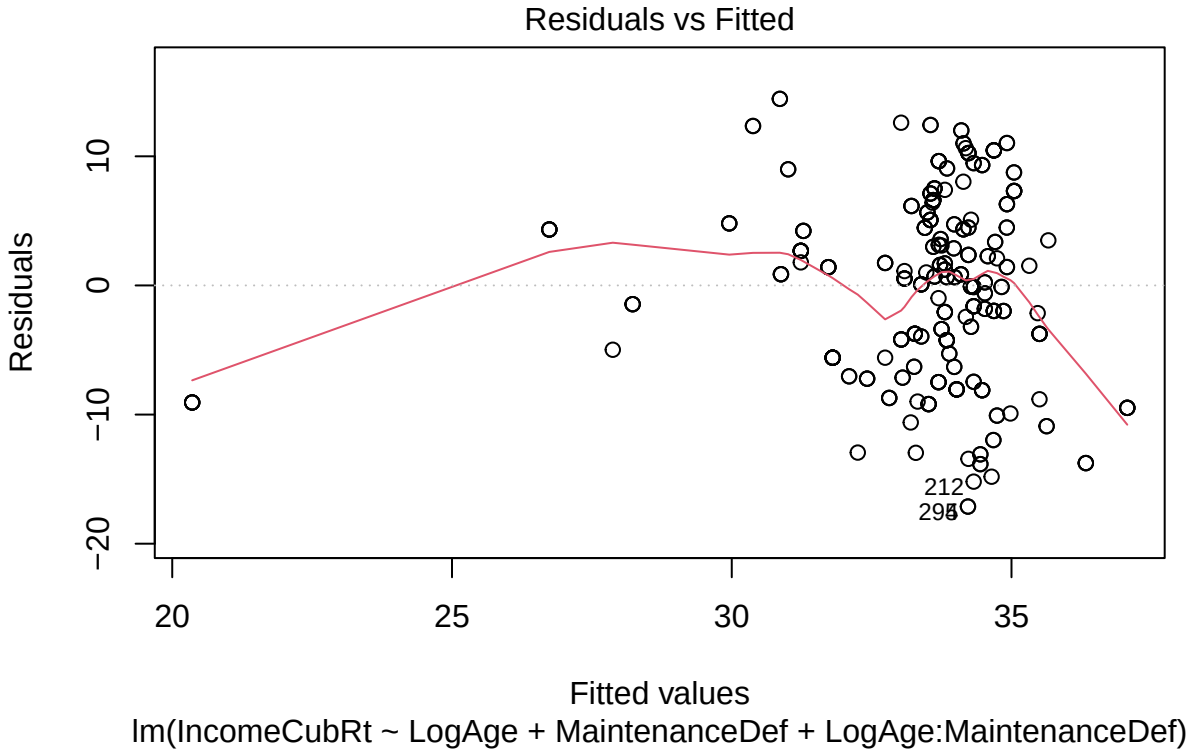
After removing `NYCMove`, the adjusted R-squared decreases slightly from 0.08462 to 0.08297, and the residual standard error increases slightly from 6.888 to 6.894. We choose the reduced model because it uses fewer predictors while still resulting in a similar fit.

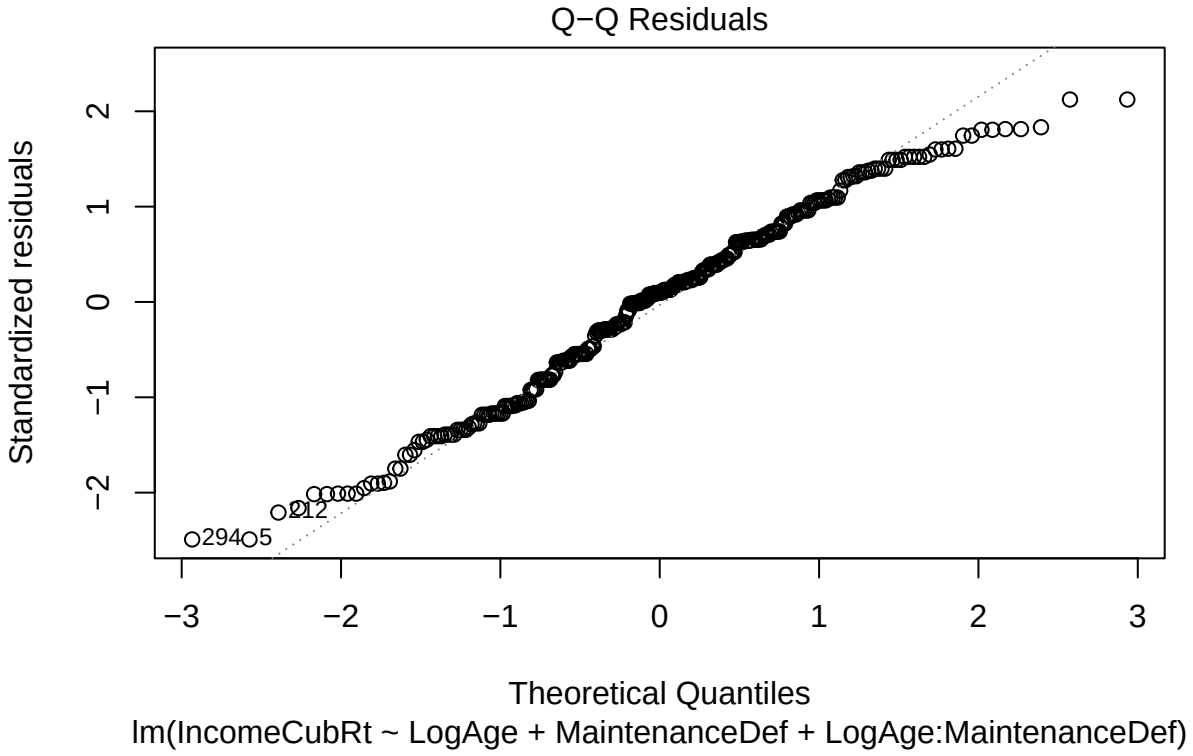
The final model is significant overall because the regression F-test p-value is about 0.00000274, which is below 0.05. The interaction term is also significant, with a p-value of about 0.0000862. The p-values for `LogAge` and `MaintenanceDef` are approximately 0.00775 and 0.0000399, so both variables are also significant at the 5% significance level.

The coefficient for `MaintenanceDef` is about -18.586, and the interaction coefficient is about 4.640. Therefore, at a given age, one additional maintenance deficiency changes the predicted cube root of household income by approximately $-18.586 + 4.640 \log(Age)$. The positive interaction coefficient means that this slope becomes more positive as log age increases. Therefore, we cannot describe the maintenance-deficiency association with total household income as being the same at every age.

The VIFs in the final model are approximately 2.67 for `LogAge`, 326.64 for `MaintenanceDef`, and 317.28 for the interaction term. These values exceed 2.5, so we cannot say that the final model has no multicollinearity concern. The high VIFs reflect the strong relationships between the interaction term and the individual variable terms. Here, we recognize multicollinearity as a potential concern.

Checking the final model assumptions





The points on the residual plot scatter around zero, but the red line still shows a curved pattern. Therefore, the linearity condition is not fully satisfied. There is no clear funnel shape, but most fitted values are crowded together and there are few observations at the lower end. Thus, the plot does not provide strong evidence that constant variance is satisfied.

The points on the Q-Q plot follow the straight line closely in the middle but deviate at both tails, particularly the upper tail. Therefore, Normality is only approximately satisfied.

Overall, the final model provides a simpler structure than the full interaction model and a higher adjusted R-squared than either model without an interaction. However, its R-squared is only about 0.09220. This means that the model explains approximately 9.22% of the variation in cube root of household income. The remaining concerns on model assumptions and low R-squared suggest that its ability to predict individual households' income is limited.

Prediction

Now we want to predict the income for a household with 3 maintenance deficiencies, and the respondent's age being 53 who moved to NYC in 1987. Since `NYCMove` was removed from the final model, the year moved to NYC does not affect this prediction.

We enter log age and the number of maintenance deficiencies directly into the prediction:

```
Predicted_Household <- data.frame(
  LogAge = log(53),
  MaintenanceDef = 3
)

Predicted_IncomeCubRt <- predict(Income_Final, newdata = Predicted_Household)
Predicted_IncomeCubRt
```

```
##        1
## 33.85442
```

The predicted cube root of household income is approximately 33.854. This is a prediction on the transformed scale, rather than income measured in dollars. We can cube this value to get a point estimate of total household income measured in dollars:

```
Predicted_IncomeCubRt^3
```

```
##        1
## 38801.29
```

The point estimate is approximately 38801 dollars. However, cubing the predicted mean on the transformed scale does not generally give the mean income in dollars. Therefore, we report 33.854 as our main prediction of the cube root of total household income of that specific household.

Conclusion

Our analysis suggests that the relationship between maintenance deficiencies and total household income varies with age. The selected model uses log age, maintenance deficiencies, and their interaction to predict cube root of household income. The significant interaction means that the association between maintenance deficiencies and income depends on the respondent's age. The year the respondent moved to NYC is not significant in the full interaction model, so we remove it to obtain a simpler model. For the household described in the prompt, the predicted cube root of household income is about 33.854, corresponding to a point estimate of around 38801 dollars.

Although the final model improves the fit compared with the models without an interaction, it only explains about 9.22% of the variation in cube root of household income. The curved pattern in the residual plot and departures from the straight line in the Q-Q plot also suggest that the assumptions are not fully satisfied, limiting our confidence in the model's predictions. The high VIFs in the interaction model also contributes to a multicollinearity concern.

There can be more analysis using additional variables such as education, employment, and number of household earners to help explain the differences in household income. Overall, such analysis can help identify patterns in income inequality and guide further research into housing challenges faced by New Yorkers.